

PRomop: A Longitudinal Decision-Ready Patient Health Record Built on OMOP CDM and FHIR R4

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DOI: [10.xxxxxx/draft](https://doi.org/10.xxxxxx/draft)

Software

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Submitted: 04 July 2026

Published: unpublished

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Summary

Caring for a patient produces a long trail of separate records: a lab result one week, a diagnosis the next, a prescription from a different clinic, often stored in different systems and different formats. Answering an ordinary clinical question — *is this patient's cancer currently responding to treatment? what was their most recent blood count? which drug regimens have they already tried?* — means collecting those scattered pieces and working out what they add up to. Today every piece of software that needs such an answer redoes that work for itself, and two programs looking at the same patient often disagree.

PRomop is open-source software that does this assembly once, in advance, and saves the result. Incoming records are stored in the OMOP Common Data Model (CDM 5.4) ([OHDSI Collaborators, 2021](#)) — the database layout used by the OHDSI observational-research community — extended with fields specific to cancer care. Like most clinical databases, OMOP spreads a patient's history across many tables, one per kind of event: all lab results in one table, all diagnoses in another, all medications in a third. That layout is excellent for recording faithfully what happened, but awkward for asking what is true about a patient *right now*, because the answer has to be reassembled from many tables every time it is needed.

PRomop adds a table called PatientRecord that holds one wide row per patient — over 300 columns — containing the current best answer for each clinical fact: disease stage, most recent lab values, current line of therapy, and so on. The row is *flattened* (or *denormalized*): facts that would normally have to be gathered by joining many tables are pre-computed and copied into a single place, so a program can simply read the answer instead of deriving it. Whenever new clinical data arrives, PRomop updates the affected parts of that row automatically. The underlying OMOP tables remain the complete historical record of everything that ever happened; the PatientRecord row records what is true now. Analytics dashboards, clinical trial matching, and treatment-guideline checking then all read the same pre-computed answers instead of each maintaining its own version of the truth.

PRomop is deployed in production across the HealthTree Foundation (14,000 blood-cancer patients) and CancerBot (3,500 patients), supporting trial matching against 6,000 actively recruiting trials across five cancer types. Checking whether a patient meets a trial's eligibility criteria runs about 37 times faster against PatientRecord than against the raw OMOP tables ([Blum et al., 2026](#)).

Statement of Need

PRomop is designed for clinical informaticists, data scientists, and developers who need to build or integrate with a longitudinal patient health record — a record covering a patient's full history over time rather than a single visit — whether to power a trial matching engine, construct feature sets for clinical ML models, or deploy patient-level clinical decision support.

Today, answering “what is this patient’s current disease status, most recent lab values, and prior therapy lines?” requires assembling and aggregating across multiple tables at query time. Every downstream application — analytics dashboards, trial matchers, CDS engines — independently reconstructs patient state: resolving lines of therapy, determining current disease status, normalizing biomarkers, reconciling conflicting source values. This re-derivation is expensive, error-prone, and a frequent source of inconsistency between applications that should agree. Neither OMOP CDM’s normalized tables nor FHIR’s resource-oriented exchange protocol (see *State of the Field*) produces a pre-computed, per-patient record ready for these use cases.

PRomop fills this gap by:

- Storing records in OMOP CDM 5.4, inheriting compatibility with the OHDSI ecosystem ([OHDSI Collaborators, 2021](#))
- Accepting FHIR R4 ([HL7 International, 2019](#)) Bundle uploads that map directly into OMOP tables (observations → Measurement, conditions → ConditionOccurrence, medications → DrugExposure + Episode)
- Automatically deriving PatientRecord via a signal chain whenever any underlying OMOP record changes, so downstream consumers never reconstruct state themselves
- Exposing a versioned REST API (/api/v1/) with an OpenAPI 3.0 schema for integration with trial matching, CDS, and analytics services
- Providing a React-based clinician interface and synthetic FHIR data generators for each supported disease, enabling fully offline development and reproducible testing

State of the Field

Several tools address parts of the problem PRomop targets, but none combines OMOP-native storage, FHIR ingestion, and a pre-computed per-patient projection in a single open-source package.

OHDSI ATLAS / HADES / ACHILLES ([OHDSI Collaborators, 2021](#); [Overhage et al., 2012](#)) are the reference tools for population-level observational research on OMOP CDM. They excel at cohort definition and epidemiological analysis but operate at the population level; they do not produce a queryable per-patient clinical state suitable for trial matching or point-of-care CDS.

TrialGPT ([Jin et al., 2024](#)) applies large language models to match patients to trials from unstructured text. It bypasses structured data entirely, trading reproducibility and auditability for flexibility. PRomop takes the complementary approach: structured, vocabulary-mapped data with deterministic derivation.

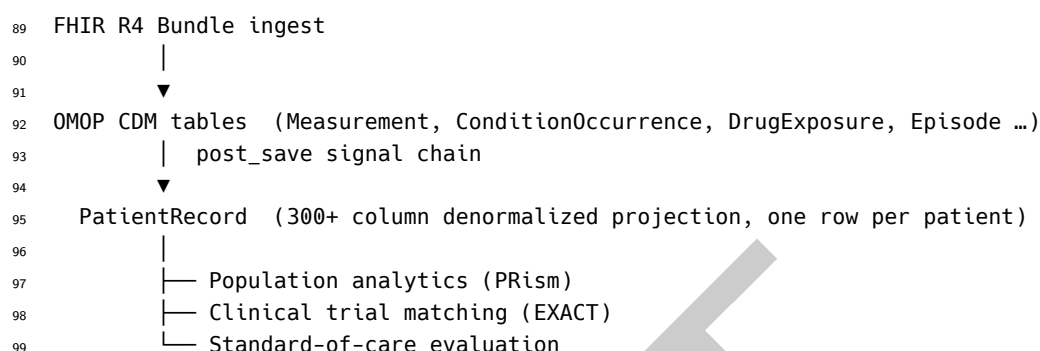
HAPI FHIR ([Smile CDR Inc., 2024](#)) and other FHIR servers store and serve FHIR resources faithfully but do not map them into a common analytical schema. Querying current patient state still requires traversing and reconciling multiple resource types.

PRomop’s contribution is the PatientRecord projection layer: a single denormalized row per patient, derived automatically from OMOP writes, that eliminates repeated state reconstruction across consuming applications.

Software Design

PRomop’s central design trade-off is **normalization for writes versus denormalization for reads**. Clinical data arrives as FHIR R4 Bundles and is written into normalized OMOP CDM 5.4 tables (Measurement, ConditionOccurrence, DrugExposure, Episode), preserving full longitudinal history and OHDSI-ecosystem compatibility. A Django post_save signal chain then derives PatientRecord — a wide, pre-computed summary table (currently over 300 columns), one

87 row per patient, storing values that would otherwise be recalculated from the OMOP tables
88 on every query — on every write.



100 This design accepts higher write cost (the projection must be refreshed on each update) in
101 exchange for dramatically lower read cost. A representative 20-criterion eligibility search over
102 raw OMOP requires 27–39 joins; against PatientRecord it is a flat predicate over a single
103 table. Benchmarks on a synthetic breast-cancer cohort measured a 37× speedup (0.30 ms
104 vs. 11.0 ms per patient) for eligibility screening (Blum et al., 2026) (??).

Table 1: Eligibility screening cost per patient (synthetic breast-cancer cohort) (Blum et al., 2026).

Approach	Joins	Time per patient
Raw OMOP, 20 criteria	27–39	11.0 ms
PatientRecord, 20 criteria	0	0.30 ms
Measured speedup		~37×

105 Key implementation components include:

- 106 ■ **FHIR-to-OMOP ingestion** mapping LOINC, SNOMED CT, and RxNorm codes into
107 OMOP tables with fallback to source value strings
- 108 ■ **Line-of-therapy inference** supplementing the ARTEMIS approach (Golozar et al., 2023)
109 with domain-specific rules for oncology regimen detection
- 110 ■ **Synthetic data generators** producing reproducible FHIR R4 Bundles per disease type for
111 fully offline development and testing
- 112 ■ **Versioned REST API** (/api/v1/) with OpenAPI 3.0 schema, OAuth2, and SMART on
113 FHIR authorization (Mandel et al., 2016)

114 Research Impact Statement

115 PRomop is deployed in production across two independent organizations — the HealthTree
116 Foundation (approximately 14,000 blood-cancer patients) and CancerBot (approximately 3,500
117 patients) — totaling roughly 17,500 patient records. These deployments support clinical trial
118 matching against 6,000 actively recruiting trials across five cancer types (multiple myeloma,
119 follicular lymphoma, chronic lymphocytic leukemia, breast cancer, and diffuse large B-cell
120 lymphoma).

121 The PatientRecord projection has enabled integration with two downstream systems: PRism,
122 a population analytics dashboard, and EXACT, a clinical trial matching engine. Both consume
123 the same pre-computed patient state rather than independently deriving it, eliminating a class
124 of inconsistency bugs between applications that previously disagreed on patient status. A
125 companion paper (Blum et al., 2026) provides architectural details and empirical benchmarks
126 of the projection approach.

127 The software is openly available, includes synthetic FHIR data generators for each supported
128 disease type, and can be deployed without access to real patient data — lowering the barrier
129 for researchers and informaticists to adopt, extend, or benchmark against their own clinical
130 data pipelines.

131 AI Usage Disclosure

132 Generative AI tools — primarily Anthropic Claude (via Claude Code) and GitHub Copilot —
133 were used extensively throughout this project. AI assisted with software development (code
134 generation, debugging, test authoring, and code review), documentation drafting (including
135 portions of this paper), and architectural exploration. All AI-generated content was reviewed,
136 tested, and edited by the author. Automated test suites (backend and frontend) were run
137 against all code changes regardless of origin. The author accepts full responsibility for the
138 correctness and scholarly integrity of the final software and manuscript.

139 Acknowledgements

140 Thank you to HealthTree Foundation for funding and feedback. Thank you to advisors Steve
141 Labkoff and Yuri Quintana for guidance and insight. The OMOP CDM vocabulary and concept
142 infrastructure is maintained by the OHDSI community. FHIR R4 specifications are published
143 by HL7 International.

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